

Mai: A Transformer Based Domain Specific Chatbot for Menstrual Health

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Traditional health-seeking behaviors, particularly in menstrual health, are deeply rooted in local languages, sociocultural norms, and religious beliefs. These factors significantly influence women in South Asia and other developing regions, often proving detrimental to their physical and psychological health. Recognizing the multifaceted nature of these challenges, this article investigates the role of generative domain-specific chatbots as a supportive mechanism to enhance menstrual health-seeking behaviors among South-Asian women. Specifically, it aims to create and assess Mai, a conversational agent, which utilizes Roman-Urdu and English models to provide accurate information, thereby contributing to the broader effort of empowering women with knowledge and resources.

Due to a lack of conversational data in this domain, our research involves creating a dataset from existing English sources and translating it into Roman Urdu, presenting its own set of challenges. These datasets, specifically curated for this research, addressed a wide range of menstrual health topics. Despite being an English monolingual model, DialoGPT's transformer architecture produced favorable results when fine-tuned on a Roman Urdu dataset. While the Roman Urdu version occasionally faced challenges with contextual understanding, both models achieved promising results. Automatic evaluation metrics indicated strong performance for Mai compared to baseline models. Likewise, user evaluations yielded average ratings of 4.2 and 4.3 (out of 5) for the Roman Urdu and English chatbots, respectively, and more than half of the medical professionals surveyed expressed willingness to recommend the dialogue system. Overall, our findings highlight the transformative potential of domain-specific dialogue systems in health education, with room for improvement in overcoming cultural and linguistic barriers to women's health information access.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; **Accessibility**; • **Computing methodologies** → **Natural language processing**; • **Applied computing** → **Health care information systems**; • **Computer systems organization** → *Real-time systems*;

Additional Key Words and Phrases: Menstrual health education (MHE), menstrual hygiene, machine learning, natural language processing (NLP), transformer models, conversational agent, generative AI, domain-specific

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chatbot, low-resource languages, Roman Urdu chatbot, multi-lingual chatbot, dataset creation, health chatbots in South Asia, stigma reduction in healthcare

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1 Introduction

In developing regions like South Asia, many women struggle to access accurate and reliable information on menstrual health due to sociocultural norms, language barriers, and limited resources. This scarcity is exacerbated in rural areas of Pakistan [22], where Urdu, despite being the 10th most spoken language globally [31], lacks sufficient technological tools to bridge this information gap. The lack of access often leads to reliance on myths and misconceptions that can negatively impact well-being, compounded by the absence of proper sanitary products [44]. Furthermore, most Asian women do not have answers to even basic questions about menstrual health [69] and cannot discuss their menstrual problems and concerns without feeling ashamed.

These taboos further reinforce the idea that menstruation, though a natural and regular occurrence for most females between ages 13 and 52 [12], remains a deeply stigmatized and forbidden topic in Pakistan. Considered a source of shame and impurity, menstruation is rarely talked about, preventing women from accessing the proper knowledge on how to manage their cycles safely and healthily [52].

Furthermore, women in Pakistan often refrain from openly discussing such issues with doctors, friends, or even family members due to societal taboos and personal shyness [59]. Research has highlighted alternative approaches to maternal health that are deeply rooted in the Islamic faith [53]. These practices, often termed as religious or spiritual therapy, typically involve consulting a spiritual healer, known as a “*pir*”—a figure of wisdom and experience—who administers blessings through the recitation of Quranic verses over consumables such as water, jaggery, or black pepper, which the expectant mother then ingests. It is also common for pregnant women to be given a “*taveez*,” an amulet inscribed with Quranic inscriptions, to wear until childbirth, believed to safeguard against malevolence and to promote a favorable pregnancy outcome, including the birth of a male heir—a preference rooted in socio-economic biases.

The endorsement of religious therapies for maternal health extends beyond specific families or pregnant women. Apart from hospital doctors, most health workers, midwives, and gynecologists at community-level health facilities advocate various forms of religious therapy, reflecting a structural problem within the healthcare environment. This systemic issue indicates that the perceptions and management of healthcare (including maternal health) among Muslims are shaped by a complex and interconnected system of sociocultural beliefs [53]. Mai, as a chatbot, aims to circumvent this issue by acting as a system of structural rebellion, instructing and educating women against the culturally established hegemonic morality.

Additionally, recent research has revealed a strong link between the belief and practice in “*Purdah*” and “*Haya*” (modesty and privacy) and how women perceive their bodies, particularly in terms of discussing and seeking knowledge about their sexual and reproductive health [54]. This connection is particularly evident in the context of Pakistani society, where women’s autonomy in health-related decisions is often compromised. In many low- and middle-income households, the prevailing family structure grants decision-making power to mothers-in-law or husbands,

which can result in women having little to no say in matters concerning their own health. Such dynamics affect various aspects of women's lives, including choices about childbirth settings, nutritional needs, and contraceptive use [54]

In various resource-limited nations such as Ghana, Tanzania, Ethiopia, Nepal, and India, extensive research has been conducted on myths and traditional practices related to menstruation. Surprisingly, the same level of attention has not been directed toward Pakistan despite its limited resources and a comparable **Menstrual Hygiene Management (MHM)** situation. Considering the nearly equal representation of females in the population, the limited research on this fundamental aspect of life is unfortunate [6].

This calls for serious efforts to increase awareness and education levels among women in both rural and urban areas of developing countries [8]. Women in this cultural context can benefit greatly from a confidential environment where they can obtain accurate information about menstruation. Such a space not only facilitates the dissemination of knowledge but also protects their identity, helping them avoid stigma and allowing them to access information freely. While some non-medicinal cultural or religious practices may not be inherently harmful [8], the issue arises when women are limited to these practices without access to scientifically valid medicinal treatments. By integrating these culturally accepted practices with evidence-based medical advice, initiatives like Mai can provide a balanced approach, respecting local beliefs while promoting effective health-seeking behaviors.

Recently, chatbots have drawn attention as virtual assistants in several domains [15]. Due to their range of applications, dialogue systems have been utilized in both commercial and non-commercial domains like online shopping assistants [19], Siri¹ by Apple, and Alexa² by Amazon. Chatbots present special opportunities to empower those in developing countries, because they can be affordable and reach millions of people [78]. They have been shown to have a positive social impact by highlighting their users' independence, efficiency, and social connectedness [30]. Thus, domain-specific chatbots designed for medical applications are key to reducing healthcare costs and improving the accessibility to medical knowledge and services for their users.

Today, **Artificial Intelligence (AI)** is being used to create conversational agents in many fields. However, not all dialogue systems are the same and can be divided into two model types: generative and retrieval-based [56]. While retrieval-based chatbots take user inputs to return the best matching from a database of predefined text messages, generative chatbots generate a response from scratch. This indicates that the dialogue is constrained and does not permit natural conversation. Due to the inherent limitation of retrieval-based chatbots, generative chatbots seem to hold the key to creating an effective conversational agent [56]. Therefore, transformers seem to be the most promising technology to create such an agent, as they have demonstrated notable gains in producing human-like language [56] and have been pivotal in advancing **Natural Language Processing (NLP)**.

To address the challenges mentioned above, our research aims to explore the potential of a chatbot designed for menstrual health, specifically targeting women who speak Urdu (the national language of Pakistan) or English and have access to smartphones in their households. Since the majority of Pakistan's population lives in rural areas where Urdu is preferred over English [14], the chatbot is designed to converse effectively in Urdu as well. This transformer-based generative conversational agent seeks to provide reliable knowledge about menstrual health while enhancing self-efficacy and maintaining user privacy. Importantly, a major contribution of this article lies in

¹<https://www.apple.com/siri/>

²<https://developer.amazon.com/en-GB/alexa>

the methodology for fine-tuning **large language models (LLMs)** to create a dialogue system that serves as an accessible resource for basic **Menstrual Health Education (MHE)**, complementing rather than replacing real-life medical consultations. To help further research, we have made the source code available on GitHub.³

1.1 Objectives

This study has three main objectives: dataset creation, chatbot, and application.

- (1) **Dataset:** The success of our research is heavily reliant on the development of a comprehensive and diverse dataset. This dataset should consist of conversations about menstruation-related topics, which is crucial for the effective training of our transformer model. Considerable effort has to be put into creating an English dataset, utilizing a range of resources to ensure its breadth and depth. This will be followed by translating the dataset into Roman Urdu, which is another major task that requires several steps to ensure correctness and no loss of content. So, a distinctive aspect of our dataset is its inclusion of both English and Roman Urdu, allowing us to create models in both languages making it more relevant to our target users.
- (2) **Chatbot:** The development of the chatbot is a central objective of our research. Our goal is to create a chatbot that can provide accurate and helpful responses to user queries about menstruation-related topics. To achieve this, we are leveraging state-of-the-art architectures such as BERT and GPT as the foundation of our dialogue system. However, we are not merely using these architectures as they are. We are customizing and fine-tuning them to suit our specific needs. This includes training them on our created dataset of English and Roman Urdu conversations. By doing so, we aim to create a chatbot that not only understands the nuances of these languages but is also well-versed in the domain-specific knowledge encapsulated in our dataset. To evaluate the performance of our models, we will use several metrics commonly used in the evaluation of LLMs. Ultimately, our aim is to create a chatbot that can provide personalized, context-aware responses to users' queries, thereby serving as a reliable source of information on menstruation-related topics.
- (3) **Mobile Application:** To give the user a way to interact with the chatbot, we will primarily be developing a mobile application that will serve as the main platform for deployment. This app will be integrated with the trained transformer models (in both intended languages) running on the back end to provide appropriate responses to the user. The mobile platform will enable us to conduct a comprehensive evaluation of the model's effectiveness through user surveys tailored to our intended audience.

1.2 Article Structure

The rest of the article has the following structure: Section 2 reviews the existing work that covers MHE in Pakistan and dialogue systems for women's health or whose theme is conversational agents using Deep Learning, primarily with Transformers. Section 3 goes over the execution of required tasks to achieve the objectives we had set, and Section 4 discusses the evaluation metrics employed to assess the performance of the trained chatbot. In Section 5, we share our findings from both automatic evaluation as well as users' interactions with the chatbot via online surveys, and field observations. Furthermore, Section 6 covers the challenges faced during this research and gives suggestions for future improvements. Last, Section 7 concludes the article.

³<https://github.com/ShafaqFatimaMughal/Kaavish-Mai-Transformer-Based-Urdu-Chatbot>

2 Related Work

2.1 Need for Combating Period Poverty and Promoting Menstrual Health Education in South Asia

Investigations into the integration of MHM within **Water, Sanitation, and Hygiene (WASH)** programs have highlighted the social and health impacts of neglecting menstrual hygiene on women, alongside showcasing successful strategies for incorporating MHM in South-Asian WASH initiatives [47]. One of the critical findings across these studies is the pervasive lack of awareness contributing to period poverty. Insights from various research indicate that in Nepal, the majority of girls receive information about menstruation from family members and peers, with only one-fifth of respondents reporting educators as their information source [1, 21, 27].

In South Asia, social taboos and rituals (religious and cultural) surrounding menstruation have led to women being excluded from various aspects of social life. This, combined with a cycle of neglect in which many women in South Asia are uncomfortable discussing menstrual hygiene in public, makes it easy for WASH initiatives to fail to include MHM as part of their initiatives [47]. For instance, in developing countries like Pakistan, women often put their faith in homeopathic and herbal treatments for menstruation-related issues, which might prove ineffective and futile [8].

Furthermore, the pervasive lack of awareness coupled with social stigmas compels women in developing regions to use unsanitary menstrual management practices, undermining health and dignity [47]. A UNICEF 2019 report underscores that achieving multiple **Sustainable Development Goals (SDGs)**—including health, education, gender equality, and water, sanitation, and hygiene—is contingent upon the prioritization of menstrual health and hygiene [55]. Moreover, inadequate and inappropriate MHM practices can lead to many issues, such as increasing one's susceptibility to urinary tract infections [61] and is associated with secondary infertility due to unsanitary MHM practices [21]. Additionally, previous studies have illuminated the psychological impact of menstruation, revealing that the initial experience can be a source of trauma and distress for girls who lack prior knowledge and understanding of menstrual health [74].

In the subcontinent (India, Pakistan, and Bangladesh), several solutions are already available to educate women about menstruation. Menstrupedia, an Indian startup, aims to fill the information gap by providing comprehensive MHE through its website and comics in several regional languages. However, it encounters challenges concerning the potential offensiveness of explicit content and a lack of active users [68]. There exist digital safe spaces in Pakistan that allow anonymous posting of content [79], but there are often issues regarding the validity and accuracy of the shared information. Consequently, there is a pressing need to address these issues and ensure the platform effectively reaches a broader audience while maintaining the utmost respect for their sensibilities and providing accurate information.

2.2 Socio-cultural Influences Affecting Menstrual Health Awareness among Pakistani Women

Despite being spoken by over 230 million people around the world, Urdu is still considered a low-resource language [31]. Research findings consistently highlight that a substantial portion of the population, particularly in Pakistan's rural areas, lacks adequate knowledge about menstruation. The language barrier exacerbates this issue, as educational materials predominantly available in English may not cater to the linguistic needs of the majority [7, 49].

Women in rural Pakistan face a double burden: disadvantaged social and economic status coupled with limited autonomy regarding their sexual and reproductive health. Culturally ingrained gender inequality restricts access to healthcare services, with women often viewed as subordinate

to men [41]. This translates to a startling statistic: Nearly half of Pakistani women have no prior knowledge about menstruation before their first period [44].

Fortunately, increased mobile phone ownership among rural women, driven by telecom initiatives, opens a potential gateway for education [48]. Our chatbot's avatar, "Mai," an experienced elder woman, leverages this opportunity. By providing information in Roman Urdu through a mobile app, Mai acts as a trusted confidante and source of accurate, culturally sensitive guidance, mimicking the traditional role of mothers and teachers as primary sources of information on menstruation [44].

Unlike unrestricted internet searches that can leave incriminating browser histories, our app provides a safe space for women to ask questions and access filtered, reliable information. This is crucial in a context where shared phones and limited personal access to devices necessitate discretion [50].

Initiatives that dismantle the language barrier, foster autonomy, and contribute to a more inclusive approach to MHE in the country are required. By emphasizing Urdu as a key language for delivering menstrual health information, we not only address the information gap but also contribute to empowering a broader population segment, ensuring a more inclusive approach to menstrual education in Pakistan. Therefore, initiatives aimed at enhancing menstrual education should strategically employ Urdu to ensure that the information is both accessible and culturally relevant to the diverse communities within the country.

2.3 Similar Domain Chatbots

Adolescentbot [59] is a menstrual health-related chatbot suggested in 2021, mainly for the women of Bangladesh. The researchers curated a dataset using local websites and social media platforms, but there did not seem to be much effort in the dataset validation. Nevertheless, Adolescentbot uses Chatterbot, a Python module that works on the similarity between the user query and reference questions stored in the dataset. If the text similarity confidence score is above a certain threshold, then the bot's response corresponding to the best-matched reference question is a reply [59]. So, in this architecture, responses are not being generated but simply mapped to an existing database. This causes the dialogue system to seem more robotic, as it does not understand the sentiment or emotion behind user queries and often fails to provide personalized responses that directly address the user's intent.

Feedpal [78], a chatbot for nursing mothers in India, mimics the retrieval-based approach of the chatbot in a single-turn scenario that only evaluates the most recent input message. It employs a database of maternal and child health questions and answers taken from a reliable book produced by a medical specialist. The bot's knowledge of the language, however, is limited. It is unable to comprehend implicit remarks and informal feedback cues. Another drawback identified by the researchers was that the history of dialogues is not considered, so it cannot process a shift in discourse automatically [78].

SnehAI [72] is an AI chatbot that addresses numerous societal taboos, such as family planning, reproductive (including menstrual), and mental health. This bot, which converses in Hindi and English, employs an NLP system to process and answer free-text queries provided by the user. This NLP system initially matches the user's intent with prefilled regular expressions, and if a match is found, then the database returns relevant responses. If no matches are identified, then the fallback policy, which is a Microsoft Language Understanding (LUIS) conversational AI service, is triggered. This service uses unique machine learning intelligence to interpret user input and respond appropriately to the bot [72].

Finally, Saathi is an Urdu chatbot that uses RASA as its core architecture and has 28 intents in its dataset. Even though it is a virtual assistant for the aged, it focuses on making general

conversations [42]. Thus, no significant research has been conducted on transformer fine-tuning for generative domain-specific Urdu chatbots.

2.4 Progress in the Latest Deep Learning Technology: Transformers

Today, there are not many “true” domain-specific chatbots, especially for MHM, as most employ Rasa, dialogue flow, or other retrieval-based technology [77]. They respond only correctly to certain inputs that the dialogue system has already seen. Thus, one cannot expect them to engage with people naturally or converse freely.

A major turning point in advancing conversational agents came in 2017, with the introduction of the transformer architecture in the groundbreaking paper “Attention Is All You Need” [71]. This new model architecture, built on Multi-head Attention Blocks and Feed Forward Neural Networks, surpassed previous sequence-to-sequence models that relied on **Long Short-Term Memory (LSTM)** or **Gated Recurrent Units (GRU)**. Transformers brought three major innovations to the field: (1) Reduced computational complexity, (2) Increased parallelization, (3) Improved ability to capture long-term dependencies across a sequence.

These advancements are central to building contextualized word embeddings, which are pivotal for transformers’ success in NLP tasks. With their deep-layered design and larger capacity, transformers have proven to be more effective than **Recurrent Neural Networks (RNNs)** in handling long-term dependencies. This capability is particularly important for tasks like conversational AI, where understanding context and nuance over extended interactions is critical [71].

Recently, pre-trained transformer-based language models have shown tremendous advances in sentiment analysis and contextual understanding across a variety of NLP tasks [65, 80]. These models are often trained to predict words based on their context on massive text data, and the learned models can be fine-tuned to adapt to various downstream tasks. Recently there has been a meteoric rise in transformer models, each with its own unique benefit.

GPT-2 is one of the most well-known auto-regressive language models and closest to our work, which learns language granularity from large amounts of open web text data. It is one of the most researched models and has many variants, including CTRL [40], Optimus [43], and DialoGPT [82]. After fine-tuning, GPT-2 significantly improves Natural Language Processing on a number of generation tasks. However, it suffers from the lack of high-level semantic control, which is paramount in language generation.

DialoGPT is developed on GPT-2 to overcome the difficulties of conversational neural response creation [82]. Thus, it is among the best options for building chatbots on generation-based models. Since DialoGPT was made to encourage more contributions, Microsoft has released the source code and pre-trained models (small, medium, and large) to facilitate future research. Even though this model works well for open-domain conversational agents, it can also be used for domain-specific agents. Furthermore, DialoGPT can be easily leveraged and adapted to new dialogue datasets, especially those with few training examples [82], making it ideal for this kind of research.

GPT-3 is a third-generation generative transformer-based model created by OpenAI to produce text from a small amount of input. The ability of GPT-3 to generate large amounts of text makes the process of creating text-based content simpler and more effective. It is typically useful for automatic human-like text generation or when having a human on hand to produce text output is impractical or inefficient [20]. Moreover, GPT-3 performs well on several tasks that call for domain adaptation, such as decoding words, inserting a new word into a sentence, or performing three-digit arithmetic [13]. However, utilizing GPT-3 for highly specific tasks might require customization and training GPT-3 on a custom dataset, which will require large computational power that will incur high costs [63].

ChatGPT is the latest large language model developed by OpenAI that works as a conversational agent. The transformer can carry long conversations and produce text indistinguishable from human speech, so much so that it has raised concerns about its potential as a tool for academic misconduct in online exams [64], among other academic areas. However, ChatGPT cannot provide accurate or up-to-date information and cannot access external information nor surf the internet (in the free version). It may not be able to generate responses to complex or unconventional questions, especially in languages other than English [24]. This inability to provide accurate information and large model footprint makes it unsuitable for our research domain.

Fortunately, recent advancements in transformer-based models offer promising alternatives with internet access capabilities. Gemini (Google AI) and Copilot (formerly Bing Chat) are prime examples. These models leverage their ability to access and process information from the web to provide more comprehensive and up-to-date responses [33, 67]. This is crucial for domains like menstrual health, where accurate information is essential for users.

However, internet access in LLMs also presents potential downsides. One concern is the potential for bias present in online information sources. These models can inadvertently inherit biases from the data they consume, potentially leading to misleading or harmful responses [10]. Moreover, these models suffer from misinformation, as the vast amount of information available online can include inaccurate or misleading sources. The models might struggle to distinguish reliable sources from unreliable ones, potentially leading to the dissemination of misinformation [51], particularly on sensitive topics like menstrual health. Additionally, the reliance on internet access raises questions about security and privacy. MHM chatbots should prioritize user privacy and ensure their queries are not used for unintended purposes [37].

Last, very little work on dialogue system development using transformers has been done in low-resource languages. Småprat, a Swedish chatbot, illustrated the potential for transfer learning of monolingual English models like DialoGPT by training it on a Swedish dataset for which it utilized Swedish language conversational datasets obtained from publicly available sources like Reddit [4].

Some recent studies on cross-lingual transferability showed that deep monolingual models could learn abstractions that generalize across languages [9]. As proven for mBERT, this goes against the prevailing opinion that links multilingual models' ability to generalize to other languages to their shared subword vocabulary and collaborative training [57]. Research on Småprat also concluded that deep monolingual models learn abstractions that generalize across languages [4]. So, this article, using Roman Urdu and English, further investigates whether this is only possible in lexically similar languages such as English and Swedish or in more diverse languages such as Roman Urdu that have some lexical similarity with English but have distinct linguistic systems.

3 Methodology

In this study, we sought to create a domain-specific Roman Urdu chatbot and comprehend its potential to enhance under-privileged women's knowledge of menstrual health in Pakistan's constrained environments. The study would help us assess how effectively a monolingual transformer model like DialoGPT can be fine-tuned on another language that uses the English alphabet (Roman Urdu in our case). On a side note, we would also be able to analyze the current beliefs of women regarding menstruation myths and taboos. The flowchart in Figure 1 represents the pipeline that we followed for this research.

3.1 Dataset

An essential element of the project is dataset creation. During the time of this research, datasets regarding our domain were relatively unavailable. Therefore, we created our datasets manually from scratch by fetching data from multiple data sources.

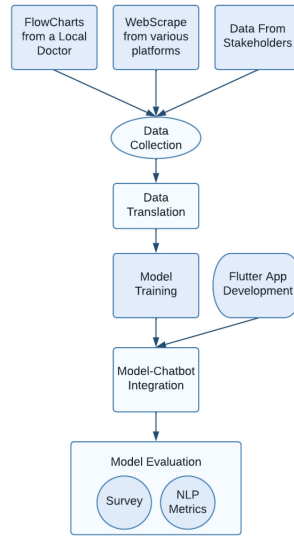


Fig. 1. Methodology pipeline employed - The flowchart shows a summary of the entire methodology opted in a step-by-step manner. The work involved data collection from various sources, dataset translation, model training, and Flutter application development, which were done in parallel. Model-chatbot integration involved creating APIs and deploying the model online, followed by evaluation using surveys and computational NLP metrics.

3.1.1 Flowcharts. The first source of the dataset was sourced from local female doctors in Karachi. These doctors work in Civil Hospital, Shaheed Benazir Bhutto Trauma Centre, and Agha Khan Hospital. The dataset was then converted into a conversational flowchart format by a certified professional from the Professional and Linguistic Assessments Board UK. She has been practicing medicine for more than eight years in Pakistan and the UK. Her experience in healthcare aided in bringing into perspective the correct medical terminologies and appropriate conversation tracks.

3.1.2 Web Scraped from Other Platforms. A lot of data was scrapped from multiple sites, including Hello Clue,⁴ Flo,⁵ and Always.⁶ This data was then manually converted from simple facts and statements to text that mimics a conversational flow between a user and an agent. To clean this data, we took a similar approach to Reference [82] by:

- Removing instances containing contacts, emails, and URLs.
- Removing instances where the target contains word repetitions.
- Removing instances containing numbers and special characters.
- Removing instances of profanity and explicit language.

3.1.3 Data from Stakeholders. After establishing a basic pipeline of transformer fine-tuning and a simple Android app was ready, we conducted a pre-pilot study with 12 female students in our institute for usability testing before releasing the prototype to a larger audience. All these students were young women, between 20 and 23 years, and smartphone users. The chat sessions with them helped us test out our application, the response time of the transformer, and the load the app could bear at a time. But, most importantly, the 12 participants filled out an online survey, via

⁴<https://helloclue.com/>

⁵<https://flo.health/>

⁶<https://www.always.com/en-us/tips-and-advice>

Google Forms, to suggest the type of conversations they would hold with the chatbot (which we will discuss in Section 5.1). This gave an idea of the type of user queries to expect. It is important to note that during the online survey, no private identifiers were required, and the forms did not collect the participants' e-mails either. We then used the data gathered from the users to construct a validated dataset consisting of user queries and ideal responses. This helped us enhance our dataset further for the next fine-tuning of the transformer model. Additionally, this pre-pilot study helped us understand user perceptions and came with an overwhelmingly positive response regarding the chatbot's user-friendliness. The feedback shared by the students also helped us improve the app's interface.

3.1.4 Translation and Transliteration of Dataset from English to Roman Urdu. All existing datasets were transliterated in two steps. First, all English data was translated from English to Urdu via Google Translate. Subsequently, the Urdu data was transliterated to Roman Urdu using a platform called *ijunoon*.⁷ While this process of translation and transliteration was largely automated through Selenium, complete automation was hindered by challenges related to gender-based words.

By default, the technologies employed above always considered the subject's gender as male. For example, the sentence "I want to help you" would be translated to "Main apki madad karna chahta hon" (the word "chahta" being a masculine verb). This posed a challenge, since the chatbot, Mai, was intended to be female. The issue was compounded by inaccuracies in the translation of certain English terminologies. For instance, "period" was taken in the time sense and translated into "mudat" (time period) instead of "mahawari" (menstruation). These had to be manually corrected. Also, since the website could not understand biological terminologies, it failed to translate such words from English to Urdu. For this, we had to ensure that once the transliterated data was available, all instances were manually proofread (before validation) to avoid discrepancies.

3.1.5 Data Validation. To enhance the validation of our dataset, we incorporated topic coherence modeling for validating the quality of our dataset. Topic coherence, a key metric in topic modeling, assesses the semantic cohesion of topics generated by a model. It leverages a reference corpus—a collection of relevant texts—to evaluate the contextual relationships between words within a topic. The goal is to achieve high intra-topic similarity (words within a topic share semantic meaning) and low inter-topic similarity (topics remain distinct). Higher coherence scores indicate more meaningful topics, enhancing interpretability and validity [60].

We leveraged **Latent Dirichlet Allocation (LDA)** to generate statistically robust topics relevant to menstrual health [11]. To ensure these aligned with the research area, we meticulously refined the initial LDA outputs through manual inspection and consultation with a domain expert by approaching our Biology Professor, who reviewed and further refined the resulting, seemingly coherent topics.

The C_v coherence model was employed for quantitative assessment of the topics using the dataset [60]. As suggested by Figure 2, all topics achieved scores exceeding 0.52. With these coherence scores, it appears that our dataset covers the most important topics regarding menstruation, including social stigmas and myths in Pakistan. So, as suggested by the heat map mentioned above, our final dataset includes the following domains:

- (1) **Periods:** Conversations about menstrual cycles, including biological processes, regularity, duration, flow characteristics, and associated physical and emotional changes.
- (2) **Infertility:** Discussions on challenges and factors related to infertility in the context of menstrual health, covering topics such as fertility, conception difficulties, and treatment options.

⁷<https://www.ijunoon.com/transliteration/urdu-to-roman>

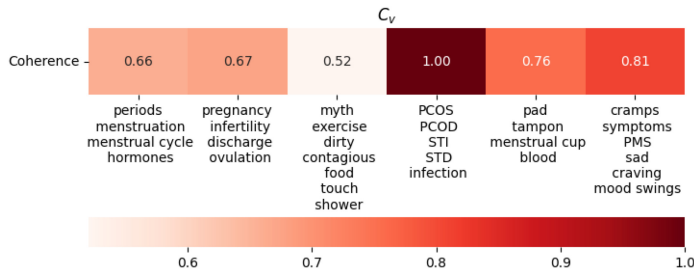


Fig. 2. A visualization of the topic coherence scores for the main identified topics that the dataset is covering. The six sets of words represent the following topics (from left to right): periods, infertility, common menstrual diseases, MHM mistakes, myths, and STDs.

- (3) **Common Menstrual Diseases:** Conversations focusing on prevalent diseases and disorders that affect menstrual health, such as **Polycystic ovary syndrome (PCOS)**, endometriosis, and other related ailments.
- (4) **Menstrual Hygiene Management (MHM) Mistakes:** Conversations about misconceptions and common mistakes in managing menstrual hygiene. Topics include appropriate hygiene practices, choice of menstrual products, and risks associated with improper management.
- (5) **Myths about Menstruation:** Conversations centered around common myths, cultural beliefs, and superstitions related to menstruation. It aims to challenge these myths and foster a more informed and inclusive understanding as well as provide accurate information and dispel stigma or misinformation.
- (6) **Sexually Transmitted Diseases (STDs):** Conversations related to STDs and their connection to menstrual health. It covers prevention, transmission, symptoms, testing, and treatment options specifically relevant to menstruating individuals.

All of this data was compiled in **comma-separated values (.csv)** files and was verified by the doctor we were consulting and the Biology Professor at our institute. Once verified, the total number of instances in our dataset .csv file was 1,917. Here, each instance consisted of two columns: user intent and chatbot response.

Since no such conversational dataset about menstruation existed, we had to create our dataset from scratch, which was a substantial part of our research. Thus, we have made both our English⁸ and Roman Urdu⁹ datasets public on Kaggle as a contribution towards future projects for social good in a similar domain. As of April 2024, there are 961 views and 93 downloads on our English dataset and 394 views and 15 downloads on our Roman Urdu dataset.

3.2 Model

Based on our research, we concluded that DialoGPT was the best model for building a generative dialogue system that understands the context of a conversation. Even though it is a monolingual model, as discussed in Section 2, research in Reference [4] suggests promising results when this model is trained on languages having lexical similarities or shared alphabets. As specified earlier, DialoGPT inherits from GPT-2 the ability to conduct a “multi-turn dialogue session as a long text and frame the generation task as language modeling” [82].

⁸<https://www.kaggle.com/datasets/shafaqfatimamughal/english-menstrual-health-and-hygiene-chatbot>

⁹<https://www.kaggle.com/datasets/shafaqfatimamughal/roman-urdu-menstrual-health-and-hygiene-chatbot>

Table 1. Hyperparameters for Optimal Model Performance

Epochs	50
Batch Size	4
Block Size	512
Temperature	0.8
Learning Rate	$5e^{-5}$

To generate a new sequence, we first concatenate all dialogues in a session into a long string $x_1, x_2, \dots, x_N$ that ends with an end-of-text token. The conditional probability of the target sentence T given the source sentence S is given by:

$$P(T|S) = \prod_{n=m+1}^N p(x_n|x_1, x_2, \dots, x_{n-1}). \quad (1)$$

A multi-turn dialogue session $T_1, T_2, \dots, T_k$, (1) can be written as essentially the product of conditional probabilities $p(T_i|T_1, T_2, \dots, T_{i-1})$ [82].

When training DialoGPT, fine-tuning the pre-trained model gives larger performance gains compared to training from scratch [82]. Since this project has a limited dataset, it is better to fine-tune an existing model. For this research, we used a DialoGPT small model with 12 transformer layers and 117M parameters. We chose this model for its balance between model capacity and training efficiency for our resource constraints environment [82]. To optimize training, we employed a grid search approach across a range of learning rates ($1e^{-5}$ to $5e^{-4}$), batch sizes (4 to 16), and training epochs (5 to 100). We minimized the **Causal Language Modeling (CLM)** loss function using the Adam optimizer. To prevent overfitting, we employed a gradient clipping with a maximum norm of 1.0. Training progress was monitored by tracking training and validation loss curves, along with perplexity on the held-out test set. The final hyperparameter configuration (shown in Table 1) resulted in the best performance based on the lowest validation loss and perplexity on the test set for both our English and Roman Urdu models. The experiments were conducted on an Ubuntu machine running an Nvidia RTX GPU with 24 GB memory, with the datasets split in a 90:10 train-test-split ratio.

In addition, we specified the conversation context as 7. Since larger contexts bring memory challenges, a context length of 7 was a good balance between performance and training [5]. However, due to memory constraints, the number of tokenized characters in our chatbot was limited to 1,024, including the conversation context.

3.3 Mobile Application

To give the user a way to interact with the chatbot, we needed to integrate our trained transformer into an application. So, we created a mobile application that integrated both the English and the Roman Urdu transformer models running on the back end. The models generate and return an output as a response to a query from the front end. These fine-tuned models were hosted on **Google Cloud Platform (GCP)**.¹⁰ We created a basic Flask REST API with a single POST endpoint that would get a reply from the deployed model if given a query.

Furthermore, we tried to create an appealing front end and a simple interface where users could log in to the application via e-mail or as guests to preserve their anonymity. We chose a color scheme that complemented the major theme of the app, as well as figures that were subtle but

¹⁰<https://cloud.google.com/gcp>

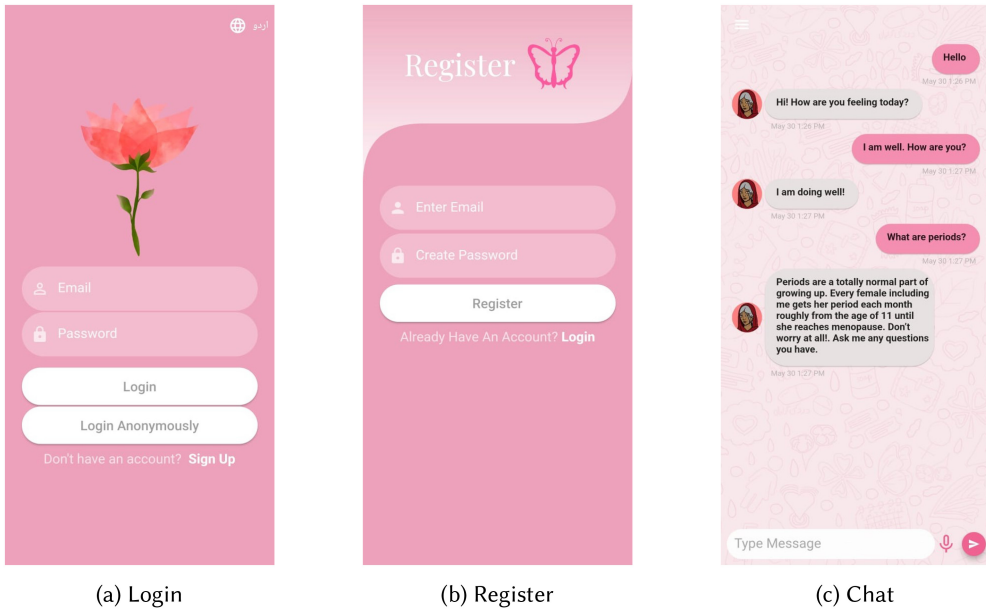


Fig. 3. Mai application screens (English) - The login screen allows users to log into the chatbot app as a user or guest. The register screen allows a female to register herself as a user. The chat screen is the main chat interface where the user interacts with the dialogue system.

complemented the theme. We also took inspiration from other MHE chatbots, including Feedpal [78], as well as the Whatsapp interface. Additionally, considering Pakistan's low literacy rate [3], we integrated text-to-speech and speech-to-text features. These enhancements cater to users who may find typing or reading challenging. Figure 3 shows the user interface of three of the main screens of the Mai application in English, while Figure 4 shows the same screens in Urdu. If a user chooses to log in, then all of their previous conversations are loaded directly from our Firebase database.

Figure 5 is a system diagram showing our application's different components. It effectively represents how the Flutter application is integrated with the fine-tuned transformer and even the Firebase database. The diagram also shows other tools and specific libraries used during the app's building or consumption.

4 Evaluation Metrics

The evaluation of our fine-tuned DialoGPT's performance was executed using the following two kinds of metrics: automatic and human.

4.1 Automatic Evaluation Metrics

Depending on the objectives and specifications of the chatbot, different metrics can be used to assess the effectiveness of a domain-specific generative chatbot. The effectiveness of NLP models, such as transformers, can be assessed using these measures. They offer a quantitative evaluation of the text produced by machines, which can be contrasted with a collection of reference texts or transcriptions. We chose to assess Mai with perplexity, ROUGE L, and BERT metrics.

Perplexity can be used to measure how well a chatbot can guess the next word in a given conversation, because chatbots are often trained on enormous volumes of conversational data. Lower

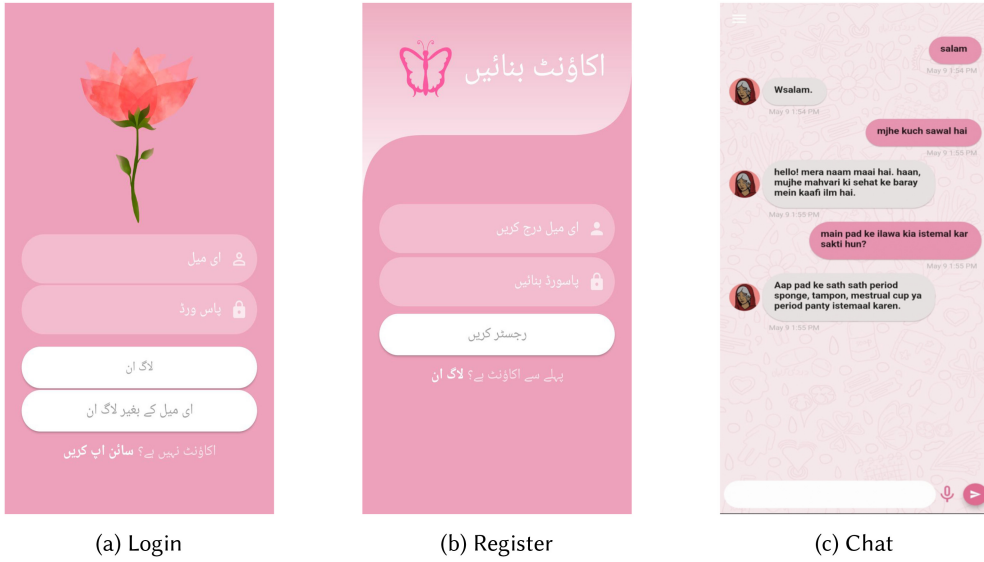


Fig. 4. Mai application screens (Urdu) - The screens are similar to the English app and show all the same screens in Urdu/Roman Urdu text.

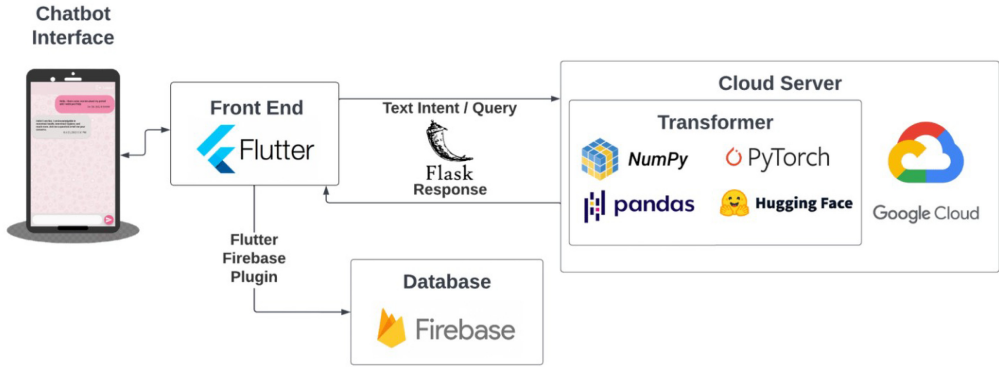


Fig. 5. System diagram of our mobile application - A high-level overview of all the main components that are part of this research project. The chatbot application interface is powered by Flutter, which utilizes a Flask API to get a response from the fine-tuned DialoGPT models that were deployed on the Google Cloud platform. The Flutter application saves these chats in the Firebase database.

perplexity ratings correspond to more accurate response generation and better language modeling [16]. This metric also correlates strongly with human evaluation [5].

BERTScore is another metric gaining traction in the evaluation of text generation models. It calculates the similarity between two sentences by summing the cosine similarities of their pre-trained BERT contextual embeddings [81]. This metric is particularly useful for dialogue systems, as it computes the F1 scores by aligning token embeddings from the human reference with the system's response, offering a nuanced measure of quality beyond mere word overlap [32]. A higher BERTScore means there is greater semantic similarity between the two texts, so a higher BERTScore is desirable [62].

ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation using Longest Common Subsequence) is a metric that can be particularly effective for evaluating the quality of text generation models like chatbots [17]. It focuses on the longest common subsequence between the generated text and a reference text, which can be a good indicator of the overall fluency and accuracy of the generated content. ROUGE-L is beneficial, because it considers sentence-level structure rather than just isolated n-grams, which may provide a better sense of the overall coherence of the generated text. It can capture the essence of the content by focusing on the longest matching sequence of words, which is often indicative of the main points being communicated. A higher ROUGE-L score indicates a longer LCS, which generally means better quality of the generated text, as it shares more similarity with the reference text. Therefore, a higher ROUGE-L score is better [17].

Furthermore, the concept of ground truth similarity is central to most automated evaluation metrics. These metrics, including the well-known BLEU and ROUGE, assess the degree to which system responses align with ground truth human responses. They typically measure this alignment through word overlap or embedding similarity, providing a quantitative measure of a system's ability to replicate human-like responses [29], which may not be the best choice. These metrics were originally designed for tasks such as machine translation and summarization. They reward responses with a high degree of word overlap to reference texts, regardless of the specific context or domain knowledge required [25]. In dialogue systems, however, there can be many good responses to a single prompt. A focus on word overlap in this context can be misleading. Additionally, research by Jurafsky and Martin [39] highlights that these methods perform poorly because of the sheer number of possible responses in open-ended dialogue.

Studies have also shown that metrics like BLEU may not correlate well with human judgments of quality in dialogue systems. High word overlap does not necessarily translate to a good response in a conversation [46]. Additionally, to have a more complete knowledge of a model's performance, it's important to remember that even automatic metrics chosen are not ideal alone and should be combined with other types of evaluation, such as qualitative analysis and human assessment.

4.2 Human Evaluation Metrics

As mentioned earlier, chatbots' effectiveness, utility, and engagement for their target users are critically dependent on human evaluation. The ambiguous and dynamic human language and communication elements help close the gap between computational evaluation metrics and actual performance. In human evaluation, test subjects can assume two primary roles: Interacting with the system and Rating dialogues [26]. To encompass both evaluative roles, we conducted dual surveys.

4.2.1 Interacting with the System. In this role, subjects actively engage in conversations with the chatbot. They experience the chatbot's responses firsthand and evaluate its conversational quality based on various dimensions [26].

The first round of surveys, which required the participants to interact with the dialogue system, was conducted across multiple institutions. This includes one university (Habib University), one school (Karachi High School), one fertility clinic (Australian Concept Fertility Centre), and one hospital (Agha Khan University Hospital) situated in different areas of Karachi, Pakistan (see Figure 6).

A total of 104 individuals participated in the survey with full consent (53 participants gave feedback on just the English chatbot that was developed earlier, while 51 participants gave it for both Roman Urdu and English). Each participant interacted with the chatbot for 7 turns, since the context length during the fine-tuning of our model was set as 7. Though we preferred in-person



Fig. 6. Pictures from various locations where the survey was conducted. They show the researchers interacting with females from all walks of life to do a fair human evaluation of the chatbot. Left: Agha Khan University Hospital, Top-right: Australian Concept Fertility Center, Bottom-right: Karachi High School.

surveys, we recognize the importance of accessibility. For those unable to participate in -person, we offered the option to complete the survey online through Forms (which is recognized as a valid form of evaluation [76]) or access the beta version of the app on the Google Playstore. This ensured inclusivity and captured valuable feedback from a wider range of individuals.

Before the survey, participants were briefed about the purpose of the study, the data to be collected, and potential risks associated with the study. To introduce them to the dialogue system, we showed them our demo videos that are uploaded on YouTube.¹¹ Importantly, participants were not informed about the internal details of the chatbot model or app; instead, a technique similar to the “Black Box” technique was employed for testing [28]. This approach ensured that participants interacted with the system without any preconceived notions about its workings, thereby allowing for a more genuine assessment of their experiences and perceptions. Additionally, our survey was conducted using Samsung A20 smartphones,¹² a mid-range option in Pakistan, to ensure uniformity and address privacy concerns related to personal devices. This standardization allowed for consistent user feedback and enabled us to evaluate the chatbot’s response time effectively.

This survey was conducted in three phases, ideally executed in one sitting. First, the users were given a Form designed to analyze different demographics and give insight into their experiences, family influences, and knowledge regarding menstruation. We then presented our chatbot app to the users, with a disclaimer of it being an underdeveloped app prone to errors. The participants

¹¹<https://youtu.be/vPf7fFop33s>

¹²<https://usa.mymobilemarket.net/product/samsung-galaxy-a20-price/>

were told that this would help test out the app and aid us in analyzing their experiences, influences, and interaction patterns. Once they conversed with Mai for 7 dialogue turns, we took feedback from the users via another Form.

For our pre-test survey, we employed a comprehensive approach to gather valuable insights into participants' understanding of menstruation, their experiences, and the influence of various factors. The survey encompassed multiple-choice questions, short answers, and open-ended inquiries. Our primary objectives were to explore opinions, elicit comments, and assess knowledge related to menstruation among the potential users. The survey focused on several key aspects:

- **Demographics:** Participants provided personal information, including age, marital status, and education level. This demographic data allowed us to contextualize responses and identify patterns across different groups.
- **Role of Influential Figures:** We investigated the roles played by family members, friends, and healthcare professionals (such as doctors) in shaping participants' knowledge about menstruation. Understanding these dynamics helped us appreciate the various sources of information and support available to individuals.
- **Addressing Myths and Misconceptions:** Our survey intentionally included questions related to common misconceptions and myths surrounding menstruation. These myths, as identified by reputable sources like UNICEF [2], can perpetuate harmful beliefs and practices. By assessing participants' awareness of these misconceptions, we gained insights into the prevalence of such harmful ideas among Pakistani women.

This aided our understanding of the current knowledge of such myths among Pakistani women.

Finally, to assess the effectiveness of the Mai chatbot, we concluded the survey with user-centric questions. Participants rated the chatbot's performance on a 5-point Likert scale [38] across key areas:

- **Comprehension:** How accurately did Mai understand your questions?
- **Information Provision:** How well did Mai provide relevant and helpful information?
- **Usefulness:** Did you find Mai's responses useful for your needs?
- **Usability:** How easy and intuitive was the Mai chatbot interface to use?

Additionally, we invited participants to share their overall satisfaction with the Mai chatbot app. Beyond the ratings, we encouraged open-ended feedback. Participants could indicate:

- **Future Usage:** Would you consider using the Mai app in the future?
- **Desired Improvements:** Are there any features or functionalities you'd like to see added?
- **Unanswered Questions:** Are there any additional questions you'd like Mai to be able to answer?

4.2.2 Rating Dialogues or Utterances. In this role, subjects assess pre-generated dialogues or individual utterances. They assign scores to various dimensions of human evaluation. This approach allows for a more focused evaluation without real-time interaction [26].

In our second round of surveys, participants rated randomly selected conversations between stakeholders and Mai from the first survey according to five dimensions: Fluency, Relevance, Coherence, Context Coherence, and Correctness [29, 73]. These metrics were chosen to ensure a comprehensive analysis of the dialogue system's conversational abilities.

For this evaluation, we engaged with 23 doctors and MBBS students as evaluators. Each participant was provided with a Form that contained two conversations between users and our Mai models, one in Roman Urdu and one in English, along with the five dimensions to be rated on a 5-point Likert scale. Three additional conversations were generated for the baseline models to be rated, using the same intents as those between user-Mai conversations. Last, the medical

Table 2. This Table Shows the Evaluation Metrics for Mai and Baseline Models

Metric	Mai English	GPT-2	Blenderbot	Mai Urdu	Seq2seq
Perplexity	1.019	1.346	1.532	1.082	1.536
BERTScore	0.992	0.857	0.823	0.881	0.798
ROUGE-L	0.997	0.859	0.846	0.901	0.812

The metrics include perplexity, BERTScore, and ROUGE-L. Lower perplexity scores indicate better performance. Higher BERTScore and ROUGE-L scores indicate better performance in terms of semantic similarity and factual correctness, respectively.

professionals were also asked whether or not they would recommend their future patients to use this app for menstrual health advice and information.

5 Results and Discussion

To create performance benchmarks for our new language model, Mai, we trained three distinct models on datasets. To establish a general baseline for English text generation, we trained both GPT-2 and BlenderBot (90M). Additionally, we leveraged a Seq2seq model to create a specific benchmark for Roman Urdu. To optimize learning, we employed a validation loss-based approach for selecting the learning rate. Training for each model continued until the validation loss plateaued, indicating no further improvement.

After extensive hyperparameter tuning, Table 2 presents the automatic evaluation metrics for our domain-specific chatbot, Mai. Perplexity scores reveal a clear advantage for Mai. Both the English and Roman Urdu versions of Mai exhibit significantly lower perplexity compared to the baselines, i.e., GPT-2 (1.346), Blenderbot (1.532), and Seq2seq (1.536). Lower perplexity indicates the model's ability to generate grammatically correct and well-generalizable responses. The results show that Mai's responses translate well to more natural and human-like interactions, as discussed in Reference [5].

BERTScore evaluation, as shown in the table, further strengthens the case for Mai's contextual understanding. By leveraging contextual embeddings, BERTScore assesses semantic similarity between generated text and reference sentences. Higher BERTScore values for Mai indicate that its responses not only align structurally but are also semantically relevant to the gold standard. This semantic alignment is critical in a domain like menstrual health, where accurate information and empathetic communication are essential. Here, Mai surpasses GPT-2 (0.857) and Blenderbot (0.823) in the English dataset, and its Roman Urdu counterpart surpasses a Seq2seq (0.881) model, highlighting its grasp of the specific domain.

Furthermore, ROUGE-L scores in the table showcase Mai's performance in preserving key information in the original text. Mai models in both languages consistently achieve higher ROUGE-L scores compared to the baselines with the following scores: GPT-2 (0.859), Blenderbot (0.846), and Seq2seq (0.812). This demonstrates Mai's ability to maintain coherence and fluency in conversations about menstrual health.

In conclusion, Mai's low perplexity scores indicate fluency, high ROUGE-L scores demonstrate coherence, and high BERTScore values showcase its understanding of semantic meaning. These attributes are crucial for Mai to effectively support and educate users on menstrual health topics within the cultural context of Pakistan.

However, according to research, intrinsic evaluation provides unclear results on the evaluation of the model [66]. Therefore, we chose to evaluate our model using human agents. The survey analysis pipeline (shown in Figure 7), inspired by the methodology in Reference [18], follows these steps:

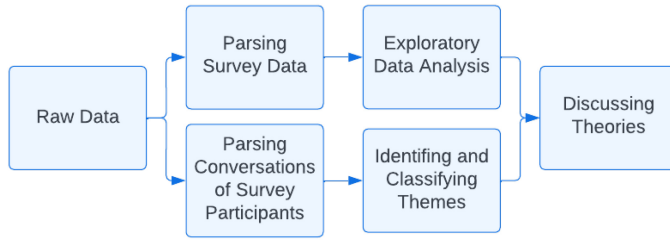


Fig. 7. Survey analysis pipeline employed - The flowchart shows a summary of how the human evaluation data was analyzed. After recording the survey data via anonymized forms and conversations with the chatbot, we parsed and organized the data. We further visualized the form data via comprehensive charts and analyzed the chats of the survey participants for errors and themes, followed by discussing some theories.

- **Raw Data:** This consists of both the survey responses and the chatbot conversations with participants.
- **Parsing Survey Data:** The raw survey data is processed to extract relevant information such as demographics, opinions, and specific feedback related to Mai’s performance. This step organizes the data into a format suitable for analysis.
- **Exploratory Data Analysis:** In this step, statistical techniques and visualizations are used to explore the dataset. Patterns, trends, and outliers in both the survey data and the conversations are identified, providing a deeper understanding of participant behavior and responses.
- **Parsing Conversations of Survey Participants:** Conversations between participants and the chatbot are parsed to extract meaningful insights. This involves structuring dialogues and identifying key points discussed by participants during their interactions with the chatbot.
- **Identifying and Classifying Themes:** Conversations are analyzed and coded based on recurring theme themes. This step organizes the findings into overarching categories, such as period discomfort, myths, and so on, which emerge from chatbot interactions.
- **Discussing Theories:** Finally, the identified themes and survey data are discussed in relation to existing theories and literature. The data is analyzed for errors and is explored for how the findings from the study either align with or challenge broader cultural and theoretical frameworks on menstrual health and hygiene.

The human evaluation of the Mai chatbot application was done via a survey conducted in different institutions, as mentioned in Section 3 (see Figure 6). The survey conducted in the three phases took approximately 10 to 20 minutes, with 8 to 10 minutes spent conversing with the chatbot.

The first Form was filled out by the 104 respondents, which allowed us to get some general demographics about our target audience to gain useful insights at the time of evaluation, as shown in Table 3. For instance, the survey respondents hail from rural as well as urban areas of Pakistan and have varying educational backgrounds. The ages of these respondents varied from 15 to 40 years, with most of them being between 15 and 25 years. These respondents mostly used pads as the main sanitary product, which consolidated the argument regarding the lack of information about menstrual products [75].

5.1 Information-seeking Behavior of the Target Audience

We examined 224 queries (124 in English and 95 in Roman Urdu) received in the conversations between the 104 participants and Mai, focusing on the nature of the information sought, themes, and framing style.

Table 3. Demographics of Survey Participants

	Responses
Total Responses	104
Highest Education	
Below Metric	1.0%
Metric/Olevel	23.1%
Inter/Alevel	19.2%
Bachelors	47.1%
Above Bachelors	9.6%
Age Distribution	
Under 15	0.0%
15–20	47.1%
21–25	28.8%
26–30	18.3%
31–35	3.9%
36–40	1.9%
Over 41	0.0%
Marital Status	
Married	39.4%
Unmarried	60.6%
Menstrual Products Used	
Pads	94.2%
Cloth	0.0%
Tampons	0.0%
Menstrual Cup	5.8%
None	0.0%
Other	0.0%

- **Query Type:** After analyzing the information-seeking habits of the users on its nature and complexity level, we represent the nature and complexity level of each query in a matrix shown in Figure 8. We found the query type to fall into two categories: fact-based and personalized broadly. 55.7% of the queries were fact-based, i.e., the questions had standard text-book answers, e.g., “When does menopause begin?”, “Kya main mahwari ke doran naha sakti hon?” (English: “Can I take a bath during the first 3 days of menstruation”). The remaining 24.5% of questions described a certain condition and seek personalized answers to a user’s health condition, e.g., “Pads mujhe raas nahin atay aur aksar kuch ghanton mein hi beh jate hain. Main kya karun?” (English: “Pads don’t suit me, and they often leak within a few hours. What should I do?”), “I am pregnant and in my second trimester, but I just had my period. What should I do?” We further divided these queries into two categories, i.e., simple and complex, based on the nature of each question. An example of a complex condition-based query is, “Main ek aur sawaal poochna chahti hoon, mujhe apne haiz ke dauran alag alag rangon ka khoon dikhaai deta hai, iska kya matlab hai?” (English: “I have another question, I noticed different colors of blood throughout my period, do they have any significance?”). Overall, our results show that 80.2% of the queries were simple in nature, showing high potential for chatbots in the domain.
- **Query Themes:** Furthermore, after researching the themes of interest to users, we discovered that period discomfort, **Premenstrual Syndrome (PMS)**, and menstrual cycles were the most common categories:

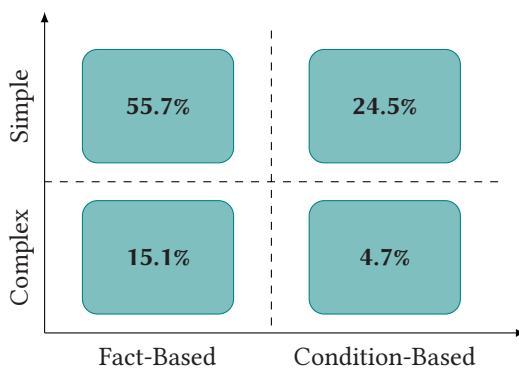


Fig. 8. User Information Seeking Behavior - The figure shows a summary of the queries received from the participants. The figure shows that Mai received different queries, where an overwhelming majority were simple queries, while a small portion was complex. This helps capture the importance of the overall usefulness of chatbots as information systems in health care.

- **Period Discomfort:** This was a prevalent topic among the queries. Users sought advice on managing cramps, bloating, and other physical discomforts associated with menstruation. Some users also asked about the normalcy of certain symptoms, indicating a need for reassurance and validation of their experiences. For example, a user might ask, “Mujhe mahwari ke doran bohat zyada dard hota hai, kya main kuch kar sakti hoon isko kam karne ke liye?” (English: “I experience severe pain during menstruation, is there anything I can do to alleviate it?”)
- **PMS:** Queries in this category often revolved around the emotional and physical changes that occur in the days leading up to menstruation. Users were interested in understanding why these changes occur and how they can manage the symptoms. A typical question might be, “Mahwari se pehle mujhe bohat zyada ghussa aata hai, kya yeh normal hai?” (English: “I get very angry before my period, is this normal?”)
- **Menstrual Cycles:** Many users had questions about the regularity, duration, and flow of their menstrual cycles. These queries ranged from general inquiries about what constitutes a “normal” cycle to more personalized questions based on the user’s unique experiences. An example of such a question could be, “Kya koi masla agar meri mahwari ka cycle bohat irregular ho?” (English: “Is it a problem if my menstrual cycle is not regular?”)
- **Infertility Issues:** Some older participants discussed with Mai their infertility issues, particularly seeking recommendations for treatments. One participant’s conversation revolved around the benefits of homeopathic medications to treat infertility, and another on depression and anxiety, due to repeated miscarriages.
- **Myths:** The survey results also showed various misconceptions and malpractices by analyzing the questions. Avoiding baths and exercise during menstruation, for example, were mentioned in the queries.

One prominent theme from the survey is the focus on period discomfort, where users sought advice on managing symptoms such as cramps and bloating. This contrasts slightly with the dataset’s broader biological understanding of menstruation, which emphasizes the menstrual cycle’s medical aspects and common diseases such as PCOS and endometriosis. While the dataset acknowledges these conditions, it lacks the user-centric perspective found in the survey, where individuals expressed a pressing need for immediate symptom relief and reassurance about their experiences.

Table 4. Sample Multi-turn Dialogue Regarding Menstruation Domain with Mai in Roman Urdu

Role	Response
User	Salam Mai. Meri tabiyat kahaarab hai aur mujhe bechaini ho rahi hai
Mai	Aap ko aisa kyun lag raha hai?
User	Mere pate mein bohat dard hai mahwari ki waja se.
Mai	Mahwari ke doran, pait mein dard hota hai jo ke neeche ke hissay mein mehsoos hota hai. Aksar auratein apni mahwari ke shuru hone se pehle ya uske pehle dino mein dard mehsoos karti hain.
User	Dard ko rokne ke liye mein kya kar sakti hoon?
Mai	Mamoolan, mahwari ke dard ka asaan ilaj hai qudrati supplements jese ke zinc, haldi, aur magnesium. Supplements dhire dhire asar karte hain, is liye inhe mahine bhar rozana lena chahiye - sirf mahwari ke dino mein nahi.
User	Main ye karungi. Shukriya.
Mai	Jab bhi aapko koi sawaal ho, mujhse dobara raabta karein.
User	Mere paas ek aur sawaal hai.
Mai	Haan, zarur poochiye.
User	Mahwari ke doran ghusl lena ke baare mein kuch uljhan hai. Kya main mahwari ke doran ghusl le sakti hoon?
Mai	Haan, bilkul le sakti hain!

Additionally, the survey's findings seem to be in line with our dataset and literature review, as the former highlights the persistence of menstrual hygiene myths and societal taboos that discourage open discussions about menstrual health. The survey revealed that users often seek not just information but validation for their personal experiences, suggesting that more research is needed in regard to the emotional dimension of menstrual health. However, while the survey did not capture conversations about STDs, these topics were notably present in the dataset, indicating a potential reluctance among users to discuss sexual health issues, possibly due to cultural stigma surrounding STDs. Another possibility is that STDs are not very rampant in Pakistan or the women did not find the chatbot to be a relevant source of help in this domain.

Myths surrounding menstruation were prevalent in our findings as well as literature in this domain. This illustrates a shared understanding of the challenges faced in combating misinformation. The survey responses and literature review both revealed persistent misconceptions, like the belief that certain activities, such as bathing or exercising during menstruation, should be avoided. The convergence of these findings across different sources solidifies the notion that these myths persist due to the misinformation, lack of education, or influence of deeply ingrained cultural beliefs, particularly in conservative contexts, which resist change [8]. Overall, these insights highlight the complex interplay between personal experiences, societal norms, and the dissemination of knowledge regarding menstrual and sexual health.

- **Query Phrasing:** The query framing in user queries varied. Many of the queries were short in length. However, they omitted essential instructive components, e.g., “My stomach hurts, what to do.” There were repeated expressions generally found in natural conversations, e.g., “I am not able to properly menstruate, how will my period flow be better, the period flow is really slow right now.”

Table 4 shows a real dialogue that shows a single conversation between the chatbot and a user during our survey (translation can be seen in Table 5). As can be seen, Mai was able to hold a multi-turn conversation easily, providing accurate MHE.

Table 5. Translated Multi-turn Dialogue Regarding Menstruation (see Table 4)

Role	Response
User	Salam Mai. My health is not the best and I feel really uneasy.
Mai	Why do you feel that way?
User	I'm on my periods and my stomach hurts.
Mai	During menstruration, it is common to feel pain in the lower part of your stomach. Most women feel pain before their periods begin or during the first few days of their periods.
User	What can I do to stop the pain?
Mai	Usually, during period you can take natural supplements that include zinc, turmeric, or magnesium. Supplements take time to show results, so take them for the entire month every day—not just during your periods.
User	I'll do that. Thank you.
Mai	If you have any more questions, you can reach out to me again.
User	I have another question.
Mai	Yes please do ask.
User	I'm confused about taking a bath/ghusl during menstruation. Can I take a bath during menstruation?
Mai	Yes you can take it definitely!

5.1.1 User Engagement. Since Mai’s avatar depicted a middle-aged knowledgeable woman and Pakistani women learn about MHM through mothers and teachers, users were comfortable in discussing menstruation-related problems and queries with the chatbot. Some associated her persona with that of a female nurse, while others saw her as a friendly and motherly figure. According to the users, the speech-to-text feature of the application was similar to those available in social media apps and made them feel as if they were conversing with a real person rather than an AI model.

The users also expressed that they preferred chatting with Mai to find answers to their queries rather than searching the internet, since it was more convenient and did not add to the browser history. This was especially applicable to women whose households had shared phones that were owned by a male family member. These women hesitated to add menstruation-related queries in their browser history and said that the Mai app icon and interface were more subtle and thus favored.

On average, the chatbot took 9 seconds to respond to a query. So, the survey participants were pleased with the chatbot’s fast response time. However, there were a few instances where the chatbot took more than this average time. While most women cooperated in these cases, some survey respondents showed impatience and annoyance. Some women perceived the cause of the slow response to be an issue in mobile data connectivity, too.

5.1.2 User Perceptions. Users expressed satisfaction with the chatbot’s ability to provide accurate and verified medical information. They appreciated the convenience of accessing the bot at any time and location, making them feel more comfortable and free from judgment. Notably, some women shared that they found the Mai chatbot to be a valuable resource, especially when they hesitated to seek advice from others. This positive sentiment from the female participants is also reflected in their responses to the second Form during the survey.

Figure 9 gives an overall feedback that the survey respondents had regarding different aspects of interacting with the Mai English chatbot. Mostly, the women were satisfied with their experience

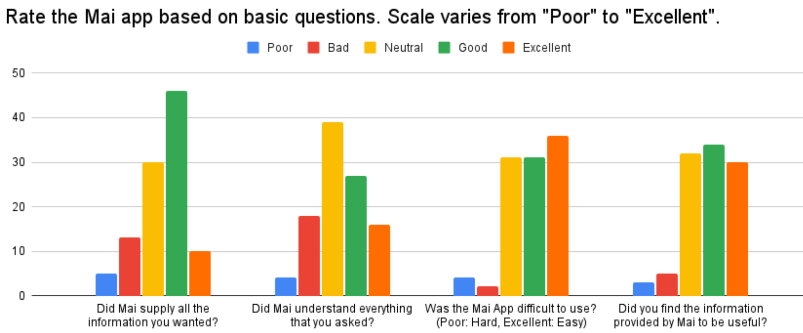


Fig. 9. Overall feedback regarding Mai English chatbot - The figure shows bar charts of the frequency of responses to questions on the feedback regarding the Mai chatbot application. The figure captures important details on information, understanding ability, and difficulty in the use of the chatbot.

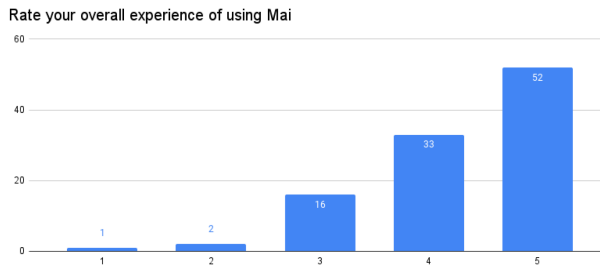


Fig. 10. Overall rating of Mai English chatbot - Out of the 104 responses, 52 gave Mai an overall rating of 5 on a 5-point Likert scale. 33 gave a rating of 4, while 16 gave a rating of 3. Only 2 participants gave a rating of 2, and 1 participant gave a rating of 1.

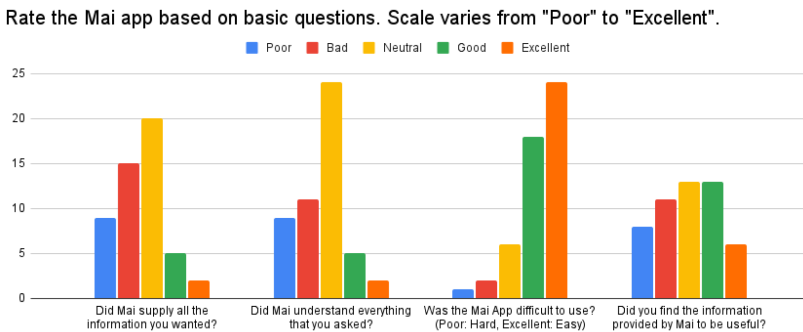


Fig. 11. Overall feedback regarding Mai Roman Urdu chatbot. Like Figure 9, the diagram shows a bar chart of the frequency of responses to questions on the feedback regarding the Mai chatbot Urdu application.

and rated the various features of the chatbot as Good or Excellent. This is even coherent with the overall rating that the women gave, as shown in Figure 10, which averages to 4.3 out of 5.

Similarly, Figure 11 gives the survey respondents' overall feedback regarding the same metrics as the English chatbot for the Mai Roman Urdu chatbot. The chatbot was unable to comprehend the question and respond with relevant facts. However, similar to the English chatbot, most women were content with their user experience and gave the app a generally positive review. The overall rating of the Mai Roman Urdu app is shown in Figure 12, which averages 4.2 out of 5. Last, the

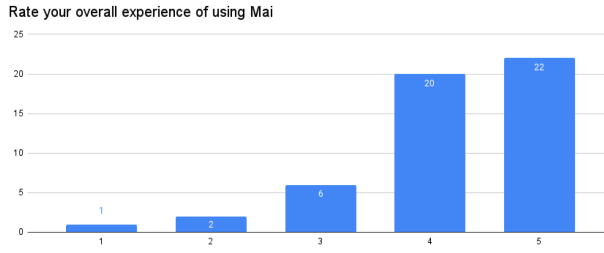


Fig. 12. Overall rating of Mai Roman Urdu chatbot - Out of the 51 responses, 22 gave Mai an overall rating of 5, 20 gave a rating of 4, and 6 gave a rating of 3. Only 2 participants gave an overall rating of 2, and with only 1 participant rating the Mai chatbot as 1.

Table 6. Feedback of All 104 Participants on the Future Usage of the English Chatbot

Response	English	Roman Urdu
Yes	67.9%	54.9%
Not Sure	25%	31.4%
No	7.1%	13.7%

Most participants chose “Yes” to use the chatbot, while a quarter replied to “Maybe.” There were very few users who chose “No.” A similar trend is seen for the Roman Urdu, where over half the participants responded as “Yes” and more than a quarter responded as “Not Sure.” Like the English chatbot model, only a few participants chose “No.”

applicability of such a chatbot in developing countries like Pakistan (where the primary language is not English) appears to be considerable. The survey results, as depicted in Table 6, suggests that respondents are open to its future use, although this does not guarantee universal acceptance or success.

5.2 Insights Derived from Survey with Medical Professionals

As mentioned in Section 4, for our second round of surveys, we engaged 23 medical professionals who rated conversations between stakeholders and Mai on five dimensions of conversations out of a score of 5. The findings revealed positive results for Mai across all measured dimensions. As seen in Table 7, compared to baseline models such as Blender-bot and GPT2, Mai English consistently achieved higher scores in Fluency, Relevance, and Correctness, and the same can be said for Mai Roman Urdu and Seq2seq model. This indicates that Mai demonstrates natural-sounding dialogue, stays focused on the topic at hand, and delivers accurate information.

For fluency in English, Mai scored slightly higher than both Blender-bot and GPT2. However, for Roman Urdu, Mai achieved a much higher score compared to Seq2seq. This suggests that Mai generates natural-sounding conversations in both languages, potentially enhancing user experience. A similar trend was seen in relevance where Mai outperformed the other models in English and similarly achieved the higher score in Roman Urdu compared to Seq2seq. This indicates Mai’s ability to stay on topic and deliver relevant responses to user queries.

Regarding coherence, Mai outperformed all other models. However, it can be seen that all three models struggled with context coherence. There is a significant drop in performance in both Roman Urdu models, suggesting further investigation might be needed to understand its root cause.

Table 7. The Table Shows the Average Score in Each Dimension of Human Evaluation Received by Mai and Baseline Models

	English			Roman Urdu	
	Mai English	Blender- bot	GPT2	Mai Roman Urdu	Seq2seq
Fluency	4.61	4.35	4.26	4.39	4.00
Relevance	4.74	4.52	4.17	4.65	4.04
Coherence	4.57	4.48	4.39	4.48	4.17
Context Coherence	4.39	4.26	4.26	4.09	3.78
Correctness	4.65	4.26	4.04	4.61	3.91

The results are given on a 5-point Likert scale given by 23 medical experts and students.

Table 8. Feedback of 23 Medical Professionals on the Future Recommendations to Patients

Response	English	Roman Urdu
Yes	56.5%	43.5%
Not Sure	39.2%	43.5%
No	4.3%	13.0

The table shows that over half the participants chose “Yes” to recommend the app to future patients, whereas only 8.7% responded as “Not Sure.” About 39.1% of the participants chose “No” to recommending the chatbot.

Finally, for correctness, Mai achieved the highest score in both languages. This highlights Mai’s ability to provide medically accurate information, a crucial aspect of an MHM chatbot.

Finally, after the survey, the medical professionals were asked about future recommendations to patients. Over half of the medical professionals surveyed indicated that they would recommend the app to their patients, further supporting Mai’s potential as a valuable tool for providing menstrual health information. Table 8 summarizes the participants’ responses regarding this.

Thus, the human evaluation survey results demonstrate that the Mai models exhibit promising conversational abilities. Mai consistently outperformed baseline models in fluency, relevance, and correctness, showcasing its potential for effective user interaction. This further reinforces the studies on cross-lingual transferability [9] and shows that deep monolingual models like DialoGPT [4] can learn abstractions that generalize across different languages that may not be lexically different.

5.3 Error Analysis

To better evaluate the limitations of the English and Roman Urdu transformer models, we analyzed their errors. We randomly selected conversations for both models in which the user gave the model a “Poor” or “Bad” rating for basic questions, as shown in Figures 9 and 11. For the English chatbot, we selected a total of 20 chats that had low human evaluation scores, while for the Roman Urdu chatbot, we selected all of the 22 chats. The results can be divided into four categories:

- **Irrelevant:** This error included responses that were outside the scope of a query or too generalized to answer the query correctly. For example, a generalized response would be something like “Getting enough sleep is important for overall health” to a query that asks, “What are some healthy foods to eat during my period?” These types of questions are hard to tackle, and the model has to be trained on a larger dataset to tackle most edge cases.

Table 9. The Table Shows a Summary of All the Conversations Rated as Poor/Bad for Both English and Urdu Models

	English Chatbot	Urdu Chatbot
Irrelevant	10	7
Illogical	7	4
Miscellaneous	5	5
Ungrammatical	0	4

- **Illogical:** This error includes responses that are contradictory or conflict with the input query. For example, in the context of activities you can do during your periods, a query that stated, “I’m feeling tired and bloated before my period. Is this normal?” to which the model responds, “There’s nothing to worry about! Exercise is a great way to boost energy levels.” These types of errors are quite difficult to handle especially when building a generative chatbot model.
- **Ungrammatical:** This error category includes the responses that are grammatically incorrect. This may be because the model fails to prevent the generation of repeated words or fails to comprehend the meaning of the words.
- **Miscellaneous:** This error type includes all other responses that are not encompassed in the above categories.

To quantify the number of errors across the models, we categorized the low-rated conversations based on the error types. The results are summarized in Table 9. This details the number of conversations in each error category for both the English and Roman Urdu chatbot models.

6 Limitations & Future Work

Transformers struggle with words outside their lexicon and thus need a lot of high-quality training data. They sometimes have difficulty finding answers that make sense in the given situation [45]. Creating a big enough dataset to meet our training requirements was difficult, so we believe that producing a larger dataset might help produce better results. Even when we managed to compile a large enough dataset in English, it was a challenge to transliterate data into Roman Urdu, considering that, currently, no platform exists that would transliterate English instances directly to Roman Urdu. Therefore, for the future, it would be beneficial to create a larger dataset for better training of the dialogue system.

Moreover, though the human evaluation survey results are promising, it is important to acknowledge that the study has limitations. Primarily, the sample size was relatively small, and the participants in the second survey were all medical professionals and MBBS students. Future research with a larger and more diverse group of participants could provide a more comprehensive picture of user perception. Another potential issue is that, since Mai is a generative chatbot and thus creates its responses to intents (based on the data it was trained on), it may provide some incorrect responses that may pose harm to the user [23].

7 Conclusion

Menstruation is a significant milestone for women, representing a social and physical transition from childhood to adulthood [34]. Despite this, the topic is stigmatized worldwide, especially in South-Asian countries like Pakistan. Women today have serious menstruation-related health concerns, and an increasing number are seeking answers online. However, most of the sources of information available in this domain are not in their local language and major form of communication—Urdu. Some other women also believe in myths or follow traditional medicine practices

(homeopathy) that are futile [8]. Thus, there is a clear need for a reliable digital solution, such as a conversational agent, that helps inform and empower its users [70] in their preferred language.

Our research addressed this gap by creating Roman Urdu and English chatbot models, integrated in a single application, specifically designed for MHE in Pakistan. Mai leverages DialoGPT's transformer architecture to provide a novel and engaging platform for vulnerable populations to access reliable MHE information. This first-of-its-kind generative chatbot effectively delivers accurate information in a non-judgmental manner, demonstrating the power of transformer models for health chatbots.

This study explored the feasibility of a chatbot in addressing menstrual health concerns within conservative societies like Pakistan, where English is not the primary language of instruction. While the Roman Urdu version faced occasional context challenges, automatic evaluation revealed that our models outperformed baseline models in perplexity, BERT and ROUGE-L scores. Further evaluations involving both medical professionals and the target audience yielded promising results. Importantly, the willingness of medical professionals to recommend the app to their patients supports Mai's value as a tool for MHE.

This strong performance can be attributed to our focus on data quality and cultural relevance. We fine-tuned the models on a comprehensive and extensive dataset specifically created for this research. This dataset covers menstrual health matters in both Roman Urdu and English, with some instances tailored to the Pakistani context. We further employed topic coherence modeling to ensure the dataset holistically addressed the major topics a menstrual health dialogue agent should be equipped to discuss with South-Asian women from diverse backgrounds. By focusing on culturally relevant information, we ensured Mai could effectively address the specific needs of its target audience.

Ethical Considerations

Ethical considerations are paramount in the construction and evaluation of dialogue systems, particularly those that address sensitive health topics. This is crucial to ensure participant well-being, data privacy, and responsible research practices. In this study, we focused on obtaining informed consent, protecting participant privacy, and minimizing any potential risks associated with discussing menstrual health online.

Upon receiving approval from Habib University's **Institutional Review Board (IRB)**,¹³ we proceeded to seek permissions from the institution's relevant authorities and began the surveys only after successful authorization. Participant confidentiality was ensured through several measures: The online surveys were designed without any personal identifiers, allowing participants to remain anonymous and promoting response integrity. Additionally, all survey questions were optional, further protecting participant privacy by allowing them to skip any questions they were uncomfortable with.

While the literature indicates a preference for traditional in-person consultations [35, 36], we recognized the cultural sensitivities surrounding discussions of menstrual health. To respect these sensitivities and promote open dialogue, we leveraged online surveys as our primary tool for engagement. This approach minimized physical interactions, thereby preserving participant anonymity and fostering a more candid exchange of information [52, 58].

However, to gain richer feedback, we complemented online surveys with in-person visits whenever possible. During these visits, participants were encouraged to share in-depth experiences and seek clarifications. We emphasized the anonymity of their contributions and assured them that the minimal data collected would be used solely for analysis. Participants also had the right

¹³<https://habib.edu.pk/office-of-academic-affairs/academic-governance-and-committees/>

to withdraw from the study at any point without consequence. Additionally, to further ensure privacy, our application was designed to allow usage without the need for registration or login, enabling users to interact as guests.

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